

* Normalization :-

Normalization is a technique of organizing the data into multiple related tables to minimize data redundancy.

Problems Occuring During Normalization:

1) Data Redundancy (means, getting same data stored in multiple databases)

2) Insestion Anomalies:

you want to insert Data:

Want to insert Data,

Data: $\left(\frac{\text{Mahak}}{\text{Name}}, \frac{170012300122}{\text{SAP-ID}}, \frac{180}{\text{Roll No.}}, \frac{19}{\text{Age}} \right)$

Table Containing:

Name	Sap_ID	Roll No
Mahak	700/2800/22	180

So, we don't have Age Column to insert the Age Value.

Hence, problem Created to insert the Age Value.
Called Insertion Anomalies.

3) Update Anomalies:

you want to Insert Data

Data: (mahak, 70012800122, 180, 19)

Now, we inserting Data & if we get a duplicated Values. Then we are Confused to update the Right Data. and we Don't know the Right Data.
 ∴ This is the Update Anomalies

4) Deletion Anomalies:

Same as Update Anomalies.
 Now, we Confused to Delete the Right Data after getting Duplicated Values

Called Deletion Anomalies.

* 1NF (First Normal Form) :- 1) States that an attribute of a table Cannot hold multiple Values. It must Hold Single Value Attribute.

Getting Data. *same*

Name	SAP-ID	Roll NO	mobile NO
Ram	700123	180	899708298
Ram	700123	180	7821026140

different

So Divide Data into Single Value Attribute

Name	SAP	Roll	mob. No	Name	SAP	Roll	mob. No
Ram	700123	180	8997082	Ram	700123	180	7821026140

* 2NF (Second Normal Form) :-

means, 1NF + No Repetition of primary key.

In short, all non-key attributes are fully functional dependent on the primary key.

eg: Data:

primary key

	Name	SAP	Roll NO	Mob. NO
1.	Ram	700123	180	8999708298
2.	Ram	700123	180	7001231891

means, we just separate the primary key dependent columns

Because, In 2NF primary key does not repeated

Name	SAP	Roll
Ram	700123	180

(2NF)

Roll	Mob.
180	8999708298
180	7001231891

(Removing from Table)

* 3NF (Third Normal Form) :-

means, 2NF + (NO) Transitive Dependency

1NF + No Repetition of primary key.

2 columns are dependent to each other.

- 1) 3NF Used to Reduce Data Duplication.
- 2) also Used to achieve Data integrity.

eg:- Data

Name	Roll No	Branch	Fees
X	21	CS	20K
Y	22	IT	75K
Z	23	CE	10K
W	24	AIML	5K

Dependent Column/
Attribute

↓ ↓
 Fees is Related to
 Branch.
 So must avoid
 the Dependency Table.

Name	Roll	Branch
X	21	CS
Y	22	IT
Z	23	CE
W	24	AIML

* Functional Dependency:

Q. R (ABCDEFGH)
F.D. (Functⁿ Dependency)

$$\left[\begin{array}{l} A \rightarrow BC, CD \rightarrow E, E \rightarrow C, D \rightarrow ABH \\ DBH \rightarrow BH, DH \rightarrow BC \end{array} \right]$$

Solⁿ:— 1) Candidate key: own milega Compulsory.

a) $A^* \rightarrow \underline{ABC}$
(No Repetition Of letters)

b) $CD^* \rightarrow \underline{CDEABH}$

Checking for Valid or not?

so. Condⁿ $\boxed{BCD \rightarrow HE} \rightarrow \text{Valid or Not}$

a) $BCD^* \rightarrow \underline{BCDAHE}$

Above Condⁿ Satisfy Because it Contains HE.

∴ Hence it is Valid.

Note:— 1) checking L.H.S of F.D. & then put up the Value of Candidate keys

Q. R (ABCDE)

F.D. $[AB \rightarrow C, B \rightarrow D, D \rightarrow E]$

Solⁿ:— 1) Candidate key:— means those L.H.S. Value get all R.H.S. Value from Relation

$$a) AB^* \rightarrow ABCDE$$

b) Candidate key: $[AB]$

Condⁿ 2:- prime Attributes: AB

(Candidate key = P.A.)

Condⁿ 3:- Non-prime Attribute: CDE

(Remaining all Non Prime)

Normalization:-

CK	P.A	Non-P.A.
AB	AB	CDE



	$AB \rightarrow C$	$B \rightarrow D$	$D \rightarrow E$	
BCNF	✓	✗	✗	→ R.H.S. CK/SK
3NF	✓	✗	✗	→ L.H.S. CK <u>OR</u> R.H.S. PA
2NF	✓	✓	✓	→ L.H.S. CK <u>OR</u> R.H.S. NPA
1NF	✓			

∴ Rⁿ is in 2NF.

* Minimal form / Canonical form:-

Q. $R(VWXYZ)$

F.D. $[V \rightarrow W, VW \rightarrow X, Y \rightarrow VYZ]$

Solⁿ:- $V \rightarrow W$
 $VW \rightarrow X$
 $Y \rightarrow V$

$Y \rightarrow X$
 $Y \rightarrow Z$

(Repeatedly Right the F.D.)

NOW, Candidate key.

- a) $V^+ \rightarrow W$ $\rightarrow$ Satisfy Cond'n $Y^+ = V$
 b) $VW^+ \rightarrow VW$
 c) $Y^+ \rightarrow Y \underline{XZ}$
 d) $Y^+ \rightarrow XZ$ (X) $\rightarrow$ out.

Final ANS:

$V \rightarrow W$
$VW \rightarrow X$
$Y \rightarrow V$

* Relational Algebra :-

1) $\sigma \rightarrow$ Select
(Cond'n)

2) $\pi \rightarrow$ projection $\rightarrow$ attributes.

Q. Output: Select tuples from Tutorial where
topic = 'Database' $\rightarrow$ Table Name.

ANS:- $\sigma_{\text{topic} = \text{"Database"}}(\text{Tutorial})$

Select
symbol

after
where
Cond'n
Write here.